# The Mean Boundary-Layer Profile

Computation of the steady laminar compressible base flow in `pyMack`

`pyMack` technical documentation

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## Contents



## 1 Scope

Linear stability theory studies the evolution of small disturbances superposed on a steady laminar base flow; that base flow, together with its wall-normal derivatives, must therefore be available before any stability problem can be posed. This document specifies how `pyMack` computes the steady, laminar, compressible flat-plate boundary layer that serves this role. Section 2 states the governing equations and the non-dimensionalisation that fixes the solver's units; Section 3 reduces the mean flow to a self-similar boundary-value problem for the profiles $\overline{U}$, $\overline{T}$, $\overline{\rho}$, $\overline{\mu}$ and describes its numerical solution; Section 4 gives the transformation from the similarity coordinate to physical, case-specific coordinates. Every equation, boundary condition, coefficient, and default value is transcribed from the implementation in `pymack/`, cited by file. The linearisation of the governing equations about these profiles is derived in the companion document *Derivation of the Disturbance Equations*; the discretisation and eigenvalue algorithms are described in *Numerical Methods*.

**Notation.** Overbars denote mean (base-flow) quantities. A prime on a mean quantity denotes $\mathrm{D} \equiv \mathrm{d}/\mathrm{d}y$, or $\mathrm{d}/\mathrm{d}\eta$ in the similarity coordinate, as stated in context. The table below collects the base-flow, transport, and scale symbols used in this document; disturbance and eigenvalue symbols are defined in the companion documents.

| | |
|---|---|
| **Base flow & transport** | |
| $\overline{U}, \overline{T}, \overline{\rho}{=}1/\overline{T}$ | mean velocity, temperature, density |
| $\overline{\mu}, \overline{\kappa}$ | viscosity, conductivity |
| $C{=}\overline{\mu}/\overline{T},\ K,\ B$ | transport groups, Eq. (11) |
| **Scales & parameters** | |
| $M_e,\ Re,\ Pr,\ \gamma$ | Mach, Reynolds, Prandtl, $c_p/c_v$ |
| $R_g,\ a_e$ | specific gas constant, edge sound speed ($a_e^2{=}\gamma R_g T_e$) |
| $L^*{=}\sqrt{\nu_e x/U_e}$ | Mack length, Eq. (7) |
| $R{=}\sqrt{Re_x},\ F$ | Mack Reynolds number, reduced frequency, Eq. (8) |
| $\alpha,\ \alpha_L{=}\alpha L^*,\ \delta^*, \theta$ | streamwise wavenumber, scaled wavenumber, displacement/momentum thickness |
| $\eta$ | similarity (generalised) wall-normal coordinate, Eq. (14) |

## 2 Governing equations and non-dimensionalisation

The flow is governed by the compressible Navier–Stokes equations, expressing conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas [1]:

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\,\nabla\!\cdot\!\mathbf{u} = 0, \qquad \text{(mass)} \tag{1}$$

$$\rho\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \nabla\!\cdot\!\boldsymbol{\tau}, \qquad \text{(momentum)} \tag{2}$$

$$\rho\, c_p\frac{\mathrm{D}T}{\mathrm{D}t} = \frac{\mathrm{D}p}{\mathrm{D}t} + \nabla\cdot(\kappa\nabla T) + \Phi, \qquad \text{(energy)} \tag{3}$$

$$p = \rho\, R_g\, T, \qquad \text{(state)} \tag{4}$$

where $\mathbf{u} = (u, v)$ is the velocity, $\rho$ the density, $T$ the temperature, $p$ the pressure, $\mu$ the viscosity, $\kappa$ the conductivity, $c_p$ the specific heat, $R_g$ the specific gas constant (plain $R$ is reserved for the Mack Reynolds number, Eq. (8)), and $\mathrm{D}/\mathrm{D}t = \partial_t + \mathbf{u}\!\cdot\!\nabla$ the material derivative. The remaining terms are the Newtonian viscous stress tensor and the dissipation function,

$$\boldsymbol{\tau} = \mu\Big(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathsf{T}} - \tfrac{2}{3}(\nabla\!\cdot\!\mathbf{u})\,\mathbf{I}\Big), \qquad \Phi = \boldsymbol{\tau}\,{:}\,\nabla\mathbf{u}. \tag{5}$$

Their closed forms are not required in what follows; it suffices that both are linear in the velocity gradients. Equations (1)–(4) are applied twice: once to obtain the steady mean flow (the subject of this document), and once in linearised form for the disturbance (the companion document).

**Non-dimensionalisation and default gas constants.** Velocities are scaled by the edge speed $U_e$, temperature by $T_e$, density by $\rho_e$, viscosity by $\mu_e$, and pressure by $\rho_e U_e^2$. Because the gas is perfect and the pressure is nearly uniform across the layer, $\overline{\rho} = 1/\overline{T}$ in these units. Three dimensionless groups govern the problem,

$$M_e = \frac{U_e}{a_e}, \quad a_e = \sqrt{\gamma R_g T_e}, \qquad Re = \frac{\rho_e U_e L_{\mathrm{ref}}}{\mu_e}, \qquad Pr = \frac{\mu_e c_p}{\kappa_e}, \tag{6}$$

where $a_e$ is the edge speed of sound. The default gas constants are the standard air values $\gamma = 1.4$, $Pr = 0.72$, Sutherland constant $S = 110.4$ K [2], and power-law exponent $\omega_\mu \approx 0.7$ when the power-law viscosity model is selected.

**The Mack length and Reynolds scales.** A flat-plate boundary layer grows as $\sqrt{x}$, so the natural local length scale at station $x$, following Mack [3], is

$$L^* = \sqrt{\frac{\nu_e\, x}{U_e}}, \qquad \nu_e = \frac{\mu_e}{\rho_e}. \tag{7}$$

With $L_{\text{ref}} = L^*$ the Reynolds number becomes the square root of the conventional one, and a fixed physical frequency $f$ maps to a reduced frequency $F$; both enter the neutral-curve and $N$-factor computations of the companion documents:

$$R = \frac{U_e L^*}{\nu_e} = \sqrt{Re_x}, \quad Re_x = \frac{U_e x}{\nu_e}; \qquad F = \frac{2\pi f\, \nu_e}{U_e^2}; \qquad \alpha_L = \alpha L^*, \tag{8}$$

where $\alpha$ is the streamwise wavenumber. $R$ is the abscissa of every neutral curve and $\alpha_L$ the ordinate. The scale conversions are implemented in `pymack/scales.py`; the solver itself is dimensionless.

## 3 The self-similar base flow

The mean flow is obtained from a boundary-value problem, not an eigenvalue problem: the data are prescribed at both ends of the domain ($y = 0$ and $y \to \infty$) and the solution is unique. Its unknowns are the steady laminar profiles $\overline{U}$, $\overline{T}$, $\overline{\rho}$, $\overline{\mu}$ and their wall-normal derivatives, which subsequently enter the linearised disturbance equations as known coefficients.

For a flat plate the mean flow is *self-similar*: profiles at different streamwise stations collapse onto a single curve in the stretched coordinate $\eta$. Introducing the dimensionless stream function $f(\eta)$ with $f' = \overline{U} = u/U_e$ (in this section a prime denotes $\mathrm{d}/\mathrm{d}\eta$), the momentum and energy equations (2)–(3) reduce under the boundary-layer approximation to the classical pair of coupled ordinary differential equations [1, 4],

$$\left(C\, f''\right)' + f\, f'' = 0, \tag{9}$$

$$\left(B\,\overline{T}'\right)' + f\,\overline{T}' + (\gamma - 1)\, M_e^2\, C\, \left(f''\right)^2 = 0, \tag{10}$$

with the transport groups

$$C = \frac{\mu/\mu_e}{T/T_e}, \qquad K = \frac{\kappa}{c_p \mu_e}, \qquad B = \frac{K}{\overline{T}} \Big( = \tfrac{C}{Pr} \text{ at constant } Pr\Big), \tag{11}$$

where $C$ is the Chapman–Rubesin parameter [5]. The final term of (10) represents aerodynamic heating: it elevates the near-wall temperature at high Mach number and couples the energy equation to the velocity field. The conductivity model is $\kappa = \mu c_p/Pr$, so that $\overline{\kappa} = \overline{\mu}/Pr$ at constant $Pr$; the transport derivatives $\mathrm{d}\mu/\mathrm{d}T$, $\mathrm{d}\kappa/\mathrm{d}T$ are evaluated analytically from the selected viscosity law.

**Boundary conditions.** Equations (9)–(10) are closed by

$$\underbrace{f(0) = 0,\ f'(0) = 0}_{\text{no slip}}, \quad \underbrace{f'(\infty) = 1,\ \overline{T}(\infty) = 1}_{\text{match the edge}}, \qquad \begin{cases} \overline{T}'(0) = 0 & \text{(adiabatic wall)} \\ \overline{T}(0) = T_w/T_e & \text{(isothermal wall)} \end{cases} \tag{12}$$

For an adiabatic wall the recovery temperature is the standard result $T_{aw}/T_e = 1 + \frac{1}{2}(\gamma-1)\sqrt{Pr}\, M_e^2$ [1]. For a cone the edge state itself is computed first, from the Taylor–Maccoll equation behind the conical shock; this is a separate calculation outside `pymack/` (see `taylor_maccoll_edge_state.py` in the Sivasubramanian–Fasel cone verification case) and is not detailed further here.

**Numerical solution.** Equations (9)–(10) with conditions (12) are solved as a five-state first-order system $[\, f, \overline{U}, \overline{U}', \overline{T}, \overline{T}' \,]$ by fourth-order collocation with residual control and Newton iteration [7], as implemented in `scipy.integrate.solve_bvp` [8], integrated to $\eta_{\max} \approx 40$, well beyond the boundary-layer edge. The driver is `pymack/baseflow.py`. On any requested grid the solver returns $\overline{U}, \overline{U}', \overline{U}''$, $\overline{T}, \overline{T}', \overline{T}''$, $\overline{\rho}$, $\overline{\mu}$, $\mathrm{d}\mu/\mathrm{d}T$, $\mathrm{d}^2\mu/\mathrm{d}T^2$, $\overline{\kappa}$, $\mathrm{d}\kappa/\mathrm{d}T$, $\mathrm{d}^2\kappa/\mathrm{d}T^2$.

Figure 1 shows the computed profiles for a representative hypersonic case: $M_e = 6$, edge temperature $T_e = 54$ K, and an isothermal wall at $T_w = 300$ K, i.e. $T_w/T_e = 5.56$. Since the adiabatic recovery temperature at these conditions is $T_{aw} \approx 7.1\, T_e$, the wall is cooled. The velocity rises monotonically to its edge value, while viscous heating produces an interior temperature maximum, $\overline{T} \approx 5.7\, T_e$, just above the cooled wall. The dotted line marks the outermost generalised inflection point, $(\overline{\rho}\,\overline{U}')' = 0$, the seat of the inviscid instability identified by Lees & Lin [6] and discussed in the companion documents.

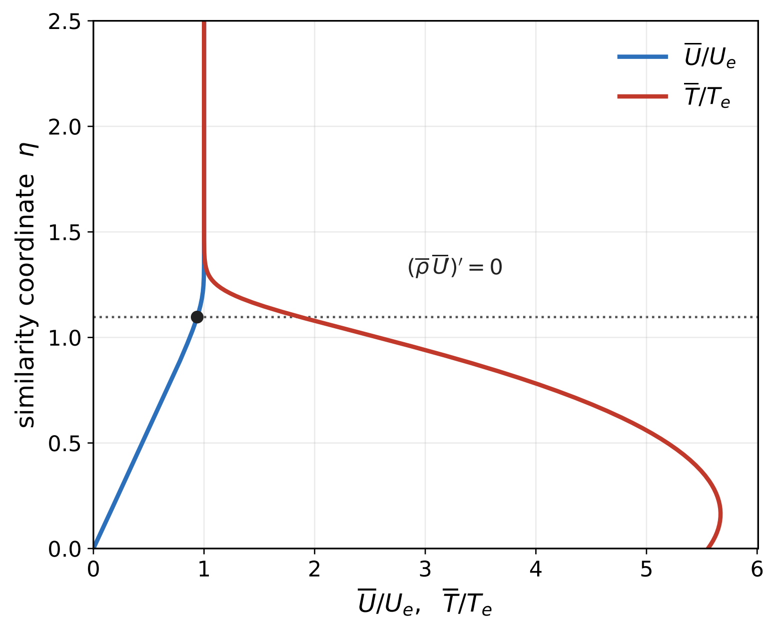


**Figure 1:** Computed compressible mean flow in the similarity coordinate $\eta$: $M_e = 6$, $T_e = 54$ K, isothermal wall at $T_w = 300$ K ($T_w/T_e = 5.56$, cooled relative to $T_{aw} \approx 7.1\, T_e$). Viscous heating produces an interior temperature maximum $\approx 5.7\, T_e$ just above the wall. The dotted line marks the outermost generalised inflection point $(\overline{\rho}\,\overline{U}')' = 0$ [6]. These profiles and their wall-normal derivatives are the known coefficients of the linearised disturbance equations; the transformation to physical coordinates is given by Eqs. (14) and (15).

**Remark: two equivalent formulations.** `baseflow.py` contains two equivalent self-similar formulations sharing the five-equation structure above. Equations (9)–(10) and the $\sqrt{2}$ transform of Section 4 correspond to the `CompressibleBlasiusProfile` form. The temporal solver and the worked example of the companion *Numerical Methods* document use `make_flatplate_profile`, which follows the transport laws and normalisation of Özgen & Kırcalı [4] ($f' = \overline{U}/\overline{T}$, factors of $\frac{1}{2}$, $y/L^* = \eta$, no $\sqrt{2}$). The two yield the same physical profiles; only the similarity bookkeeping differs.

## 4 Transformation to physical coordinates

The similarity solution is computed once, in $\eta$; all case-specific dimensional information enters only through the transformation back to the physical wall-normal coordinate. Because the near-wall gas is hot ($\overline{T} > 1$) and correspondingly rarefied ($\overline{\rho} = 1/\overline{T}$), equal increments of $\eta$ span unequal increments of physical distance. The map follows directly from the temperature field. In the units of Section 2 the differential relation is

$$\frac{\mathrm{d}y}{\mathrm{d}\eta} = \sqrt{2}\,\overline{T}(\eta) \qquad (y \text{ in units of } L^*), \tag{13}$$

whose integral gives the physical coordinate at station $x$,

$$\frac{y}{L^*} = \sqrt{2}\int_0^\eta \overline{T}(\eta')\,\mathrm{d}\eta'. \tag{14}$$

Substituting the Mack length (7) yields the fully dimensional, case-specific form

$$\boxed{y^*(\eta; x^*) = \sqrt{\frac{2\,\nu_e\,x^*}{U_e}}\int_0^\eta \overline{T}(\eta')\,\mathrm{d}\eta'} \tag{15}$$

so that a single similarity solution generates the entire family of dimensional profiles, parameterised by the streamwise station $x^*$ and the edge state $(U_e, \nu_e)$. The map (14) is strictly monotone, and its inverse assigns a unique $\eta$ to any physical height $y^*$. Since $\overline{T} > 1$ near the wall, the transform outpaces the linear reference $y/L^* = \sqrt{2}\,\eta$ there: a uniform $\eta$ mesh maps to a physical mesh stretched into the hot near-wall region, which is why $\eta$, rather than $y$, is the natural computational coordinate. Wall-normal derivatives convert by the chain rule implied by (13),

$$\frac{\mathrm{d}}{\mathrm{d}y} = \frac{1}{\sqrt{2}\,\overline{T}(\eta)}\frac{\mathrm{d}}{\mathrm{d}\eta} \qquad (y \text{ in units of } L^*). \tag{16}$$

The integral thicknesses follow from the same transform,

$$\frac{\delta^*}{L^*} = \sqrt{2}\int_0^\infty (\overline{T} - \overline{U})\,\mathrm{d}\eta, \qquad \frac{\theta}{L^*} = \sqrt{2}\int_0^\infty \overline{U}\,(1 - \overline{U})\,\mathrm{d}\eta, \tag{17}$$

where the displacement thickness $\delta^*$ measures the outward displacement of the outer flow by the layer and the momentum thickness $\theta$ its momentum deficit. A profile may be stored on a $y/\delta^*$ or a $y/L^*$ grid; the constant $\delta^*/L^*$ converts between them, with $\mathrm{D} \mapsto (\delta^*/L^*)^{-1}\mathrm{D}$. These conversions are implemented in `pymack/scales.py`.

The profiles $\overline{U}, \overline{T}, \overline{\rho}, \overline{\mu}$ and their wall-normal derivatives assembled here are precisely the coefficients of the linearised disturbance equations derived in the companion document *Derivation of the Disturbance Equations*.

## References


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